



How to Use Shutter Speed?

Get to know your camera. How to use the shutter speed to get effects such as freeze or motion blur
<http://dgit.in/SXQ1F4>

Tesla museum, a reality

A museum dedicated to Nikola Tesla hit its \$850,000 goal in less than a week on the internet courtesy The Oatmeal
<http://dgit.in/NJD00j>

Projector Test

HOW WE TESTED

Since we deal with variable light sources on the different projectors, a bit of prep work is needed to ensure uniform test conditions. We tested the projectors in a dark room with no reflective surfaces (glass, mirrors, etc.). A high gain white screen measuring 6-feet-by-4-feet (with a black background) was suspended 2 metres from where the projectors were kept on a table at a height of 3 feet from the floor. We used a Minolta autometer IV light meter to measure the light coming from the projectors – all testing was done after ensuring that there was next to no ambient light in the room (before firing up every projector, we ensured the meter showed E or Error, which means not enough light to register a reading).

Performance: We hooked up all the projectors to a standard rig with an HDMI port. The test projectors were connected to the rig via an HDMI cable to run video files and benchmarks off it. To ensure uniformity,

all projectors were restored to factory settings before benchmarking their individual performance. Prime among the synthetic tests was DisplayMate's suite of industry standard display benchmark (for more info visit www.displaymate.com). The program lets you test a projector's image for various things such as distortion, sharpness, brightness, colour, pixel density, etc. It's a comprehensive test, used to test not only projectors but TVs and monitors, and we've included only a small snapshot of it in the comparison table.

Brightness was calculated by displaying a full white image and taking a reading at the screen's surface, with the light meter's sensitive bulb facing the full glow of the projector's uninterrupted light. We also ran PixPerAn's readability test to measure ghosting and text clarity.

Apart from these objective tests, we watched movies such as *Avatar*, *Batman Begins*, *Up*, *Transformers*, *Sin City* and *Quan-*

tum Of Solace – all HD 1080p MKV files. We looked for ghosting, inconsistencies in colour, colour flashes and rainbow effects which usually occur while watching fast-paced, colourful cinematic visuals. For the projectors sporting onboard speakers, we tested their sound quality, too. We also checked each projector's startup time and its peak operating temperature (near its exhaust vent) through a digital thermometer.

Features and Design: Parameters to differentiate between the projectors – like type and number of connectivity ports, native resolution, warranty, etc., and specifications that didn't have a direct impact on their performance – were tabulated under Features. The projector's build quality, ease of setup, quality of on-screen interface, and other similar salient attributes were tabulated under Design. After applying our formula to determine winners, we came up with the Best Performer, Best Buy and Editor's Pick.

picture was a hint of moiré and banding. Also, it isn't as sharp as the EH-TW3600 projector, but still sharper than most. The Epson EH-TW6000's display of colour reproduction, skin tones, hues and contrast levels was very good, though, and it performed best in our text readability tests.

First of a few portable LED projectors in this test, the LG HW300G isn't very bright, but reproduces colours well. However, colours appear a tad muted and toned down because of its

lack of adequate brightness. The LG projector also suffers from moiré patterns, especially on a predominantly white scene.

The second Casio projector, the XJ-A240V, has good onboard audio. But in terms of display performance, the projector lacked enough blue, compensating its loss by excess pink and green tinges in lighter scenes. Rainbow effects were revealed, too. For e.g., in *Avatar*, the Na'vis lacked blue and the surroundings were touched with a hint of pink. In *Batman Begins*,

rainbows were openly visible. In *Up*, pinks were exaggerated, despite sharp and bright images and warm skin tones. The Dell M110 is surprisingly good and efficient in

its performance, despite its portable and anorexic dimensions. The projector isn't very bright but it has a very good contrast ratio, not too mention a solid performance through our DisplayMate tests. When it comes to movie performance, the Dell isn't just better than any other portable projector but beats bigger and more expensive projectors in the lineup. It does exaggerate blues a bit, but also delivers sharp, crisp images with very good contrast.

The Portronics POR 315 and BenQ Joybee GP2 are strictly portable projectors and not strong contenders for your home multimedia setup, we found out in this test. Not only did these two projectors have the poorest brightness and DisplayMate scores, they just weren't good enough for watching movies – visuals were a mishmash of moiré, rainbow effect, banding, and ghosting.

Winners

For its superior visual performance both in objective benchmarks and subjective tests, the **Epson EH-TW3600** bagged the **Best Performer** award – the Viewsonic pro8200 and Epson 6000 were second and third, respectively. Although it didn't come with a remote control and despite being a portable projector, the **Dell M110** bagged our **Best Buy** award for giving the best bang for your buck by delivering a combination of good performance and healthy features. For the lowest price, it meets our minimum standard of what a basic multimedia projector should provide. Finally, the **Viewsonic pro8200** rightly claimed the **Editor's Pick** award for being the brightest projector of the bunch, performing very well and supporting a plethora of features, it's a worthy recommendation for any home entertainment system. **1**



Dell M110